

Thermodynamics HW 3:

1. a. $p = 100 \text{ lbf/in}^2$ $v = 510 \text{ ft}^3/\text{lb}$ water

$T =$

$v =$

$h =$

0%

b. $T = 40^\circ\text{F}$ $x = 0.5$

ammonia

$p = 73.359 \text{ lbf/in}^2$

$v = 3.964 \text{ ft}^3/\text{lb}$

$u = 622.30$

X

33%

c. $p = 80 \text{ bar}$ $v = 0.0034 \text{ m}^3/\text{kg}$ water

$T = 480^\circ\text{C}$

$u = 3025.7 \text{ kJ/kg}$

$h = 3348.4 \text{ kJ/kg}$

✓

100%

d. $T = -20^\circ\text{C}$ $h = 235.31 \text{ kJ/kg}$

$p = 1.3299 \text{ bar}$

$v = 0.1464 \text{ m}^3/\text{kg}$

$u = 215.84 \text{ kJ/kg}$

✓

100%

2. $T_1 = 180^\circ\text{C}$ $T_2 = 100^\circ\text{C}$

$v_1 = 0.1941 \text{ m}^3/\text{kg}$

$p_1 = 10.02 \text{ bar}$

$p_2 = 1.01 \text{ bar}$

$0.144 = 0.0014 + x(1.67 - 0.0014)$

$x = 0.15$

✓

100%

3. phase 1:

~~$V_1 = 0.606 \text{ m}^3/\text{kg}$~~

~~$V_1 = 1.2116 \text{ m}^3$~~

~~$T_1 = 133.6^\circ\text{C}$~~

phase 2:

~~$V_2 = 1.032 \text{ m}^3/\text{kg}$~~

~~$V_2 = 2.064$~~

~~$T_2 = 133.6^\circ\text{C}$~~

$$W = -p\Delta V$$

$$W = -(3 \text{ bar})(2.064 - 1.212) = -258 \text{ kJ}$$

4. phase 1:

$$p_1 = 16 \text{ bar}$$

$$T_1 = 60^\circ\text{C}$$

$$v_1 = 0.0231 \text{ m}^3/\text{kg}$$

$$V_1 = 0.0231 \text{ m}^3$$

$$u_1 = 268.35 \text{ kJ/kg}$$

phase 2:

$$p_2 = 2 \text{ bar}$$

$$T_2 = 30^\circ\text{C}$$

$$v_2 = 0.1187 \text{ m}^3/\text{kg}$$

$$V_2 = 0.1187 \text{ m}^3$$

$$u_2 = 253.1 \text{ kJ/kg}$$

$$W_2 = 20 \text{ kJ}$$

$$Q_{12} = ??$$

0/6

5. $m = 1.50 \text{ kg}$ (liquid-vapor)

$$T_1 = 70^\circ\text{C}$$

$$T_2 = 120^\circ\text{C}$$

$$p_1 = 31.14 \text{ bar} = 31.2 \text{ kPa}$$

$$p_2 = 1.985 \text{ bar} = 198.5 \text{ kPa}$$

$$v_{f1} = 0.00103$$

$$v_{g1} = 5.042$$

$$v_{f2} = 0.00106$$

$$v_{g2} = 0.8919$$

$$0.8919 = 0.00103(5.042 - 0.00103)$$

$$x = 0.18$$

$$h = 293 + 0.18(2338 - 292.98)$$

$$h = 660 \text{ kJ/kg}$$

$$h_2 = 2202 + 803.71 = 2706.31$$

$$Q = m(h_2 - h_1)$$

$$Q = 1.50(2706 - 660) = 3069 \text{ kJ}$$

80%

6. a. $p = 10 \text{ bar}$ (liquid-vapor) $\Rightarrow V_g = 0.02362$
 $m = 25 \text{ kg}$ $V_f = 0.00084$
 $x = .6$

$m_g = 37.5 \text{ kg}$

$V = (-0.00084) + .6((0.02362) - (0.00084))$
 $V = 0.015 \text{ m}^3/\text{kg}$

$V = .015(37.5 + 25 \text{ kg}) = .906 \text{ m}^3$

b. $\frac{0.866}{0.906} = .956$

7. $m = .2 \text{ g} = .0002 \text{ kg}$
 $T_1 = 27^\circ\text{C} = 300 \text{ K}$
 $p_1 = 1.3 \text{ bar}$

$T_2 = 17^\circ\text{C} = 290 \text{ K}$
 $p_2 = 0.80 \text{ bar}$

a. $R = \frac{8.314}{4.003} = 2.077 \text{ J/g}\cdot\text{K}$

b. $V_1 = \frac{(2.076)(.0002)(300)}{1.3 \times 10^5} = 0.000557 \text{ m}^3$

c. $V_2 = \frac{(2.076)(.0002)(290)}{1.3 \times 10^5} = 9.26 \times 10^{-11} \text{ m}^3$

80%

80%

$$8. T_1 = 25^\circ\text{C} = 298\text{K}$$

$$P_1 = 2.75 \text{ bar}$$

$$V_1 = 0.05 \text{ m}^3$$

$$T_2 = 257^\circ\text{C} = 530\text{K}$$

$$Q = 2.75 \text{ kJ}$$

$$n = \frac{P_1 V_1}{RT} = \frac{(2.75 \text{ bar})(0.05 \text{ m}^3)}{(8.314 \text{ J/mol}\cdot\text{K})(298\text{K})} = 0.0055 \text{ mol}$$

(8)

$$n = \frac{m}{M} = \frac{0.24 \text{ kg}}{44 \text{ g/mol}} = 5.45$$

$$W = Q - n(u_2 - u_1)$$

$$W = (2.75) - 5.45(14.6 - 6.8)$$

$$W = 15.8 \text{ kJ}$$